FINTECH / AI IN FINANCE — HACKATHON PROJECT

DALIL

Evidence-grounded AML investigation copilot

Local LLM, no external AI API · synthetic data · the analyst — not the model — decides

`SC-01..06` `[E1] -> evidence` `sha256 verify` `TEMPLATE MODE`

Live demo: **https://cornbread-left-licking.ngrok-free.dev**

MKBANK FCC — proposed pilot client (partnership not confirmed) · data: synthetic CC0 casebook

The problem: alerts nobody has time to read

40

alerts per analyst a day (assumption)

30-40 min

per alert (assumption, not measured)

57%

false positives — rules-only layer on our own synthetic benchmark*

Why not just call a cloud LLM?

- Banks restrict sending customer transaction data to external AI services
- Every claim in a SAR-ready note must be traceable to the underlying evidence
- General-purpose models can't show which transactions their sentences came from

Rules don't just drown analysts — they still miss 20.9% of novel laundering variants (our synthetic novel split).

* measured on our held-out synthetic test set (novel split: 58.1%) — a baseline for the pain, not a bank statistic.

DALIL: one screen from alert to decision

1 0-100 score

Calibrated GBM (70%) + transparent scenario rules (30%), banded low / medium / high / critical

2 Typology hits

SC-01 structuring .. SC-06 turnover mismatch — each hit linked to its transactions

3 Evidence note

Template note built only from evidence [E#]; optional experimental local-LLM draft

4 Machine checks

[E#] exists · numbers & dates match cited evidence · meaning not checked — warnings

5 Attribution

Leave-one-out: which features moved the score, with sign and magnitude

6 Audit + export

Decision bound to analysis_id + input hash; server-built package, per-section SHA-256

One host, local inference — no external AI API

single host (demo: laptop)

CSV / alert queue
jury input or demo case

→

Ingest
strict: bad rows rejected

→

Entity resolution
ambiguity-guarded merges

→

49 features
analyst-readable names

→

SC-01..06 + GBM
rules 30% · model 70%

→

Ensemble score
band 0-100

Evidence pack
E1..En from features + txs

→

llama.cpp :8081
local GGUF, OpenAI-compatible

→

Note checks
IDs · numbers · dates (warn)

→

Decision log
analysis_id + hashes

→

Export bundle
canonical JSON + sha-256

SSE stage stream -> React SPA: queue / investigate / system — stages stream live while the pipeline runs

No external AI API · outbound traffic not audited · the ngrok demo runs on the presenter's laptop with synthetic data — a bank rollout needs its own deployment, security review and model-risk validation.

LIVE DEMO

100 seconds — real UI, real pipeline, no edits

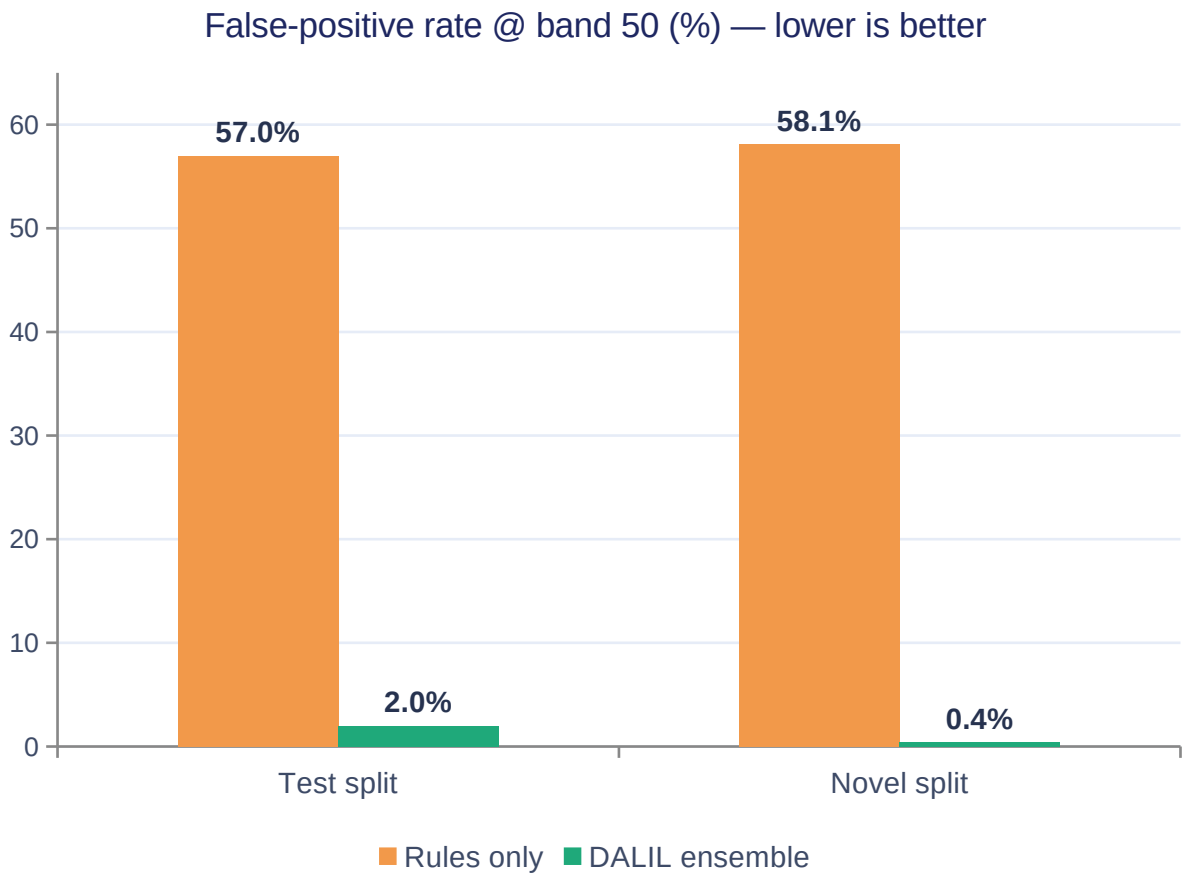
1. New CSV -> load template (12 synthetic txs, no customer metadata) -> Run
2. Score 83 critical · SC-01 hit · SC-03/SC-06 NOT EVALUATED · PROVISIONAL
3. Click a citation [E#] -> source transaction highlighted and scrolled into view
4. Separate checks: IDs exist / numbers & dates match / meaning NOT machine-verified
5. Name + justification -> Escalate -> reload (from server) -> Export: integrity + server match

`tunnel down -> localhost:8000`

`LLM down -> template note only`

`edited file -> scripts/verify_export.py (CLI)`

Honest numbers — synthetic, held-out, reproducible



99.2%

ensemble recall @ novel split, with 0.4% FPR — held-out synthetic variants

96.4%

precision @ 10% ASSUMED prevalence (novel) — a planning assumption, not bank data

0/20

LLM drafts with zero check issues — numbers match cited evidence 20.0%; meaning not measured

189 ms / 19 s

pipeline p50 (no LLM) / experimental LLM draft p50, 4 CPU threads

ROC-AUC 0.9945 / 0.9937 (test / novel) is optimistic because the data is synthetic. Failure list (5 FPs, 1 FN) ships in the UI. Full report: eval/results/eval_report.json

What is checked — and what is not

Citations exist in the evidence pack

Every [E#] must resolve to a real evidence id; a note with zero citations is reported as 0, not 'N/A'

Numbers & dates match cited evidence

Amounts (incl. 'mln'), counts and dates must match the evidence cited in that sentence — else flagged. Warnings, not a block.

Semantics NOT machine-verified

A valid citation does not make a claim true — the analyst reviews meaning; the LLM draft is experimental

Verifiable export

One JSON package:

input + SHA-256 · analysis_id · model/ruleset ids · results · evidence · note + LLM draft · decision

Built by the server from its own record; integrity (hashes) and server match are checked separately.

Limits printed inside the file: a hash is NOT a signature; an attacker holding the whole package can rewrite content and hashes together; LLM text is not bit-reproducible.

Auditable by design — and by omission

What ships in the repo

- Synthetic casebook: 3 880 cases, CC0, generator + DATA_CARD committed
- Code: MIT · model: Qwen2.5-1.5B (Apache-2.0) · llama.cpp (MIT)
- Ground-truth labels exist only offline — the API strips them from /api/queue
- 91 pytest tests + browser E2E: strict input, RU/UZ ambiguity, LLM down/truncated, decision binding, forged results, export tamper

What we deliberately did NOT do

- Copy work/tahlil-reference — it ships without a LICENSE, so no code, data or weights were reused
- Merge its standardized operation-size indices into our UZS pipeline — that would silently corrupt amounts
- Claim 'no hallucinations', production-readiness, or measured ROI — none of those are true today
- Hide failures — 5 FPs / 1 FN are published in the System tab, and every assumption is labelled

Economics: cheap to run, honest about assumptions

$0

external AI API fees per case — local open weights. Hardware (≈ $55/mo) and staff still cost: $0 ≠ free.

$30k / mo

modelled saving — ASSUMPTION: 4 000 alerts × 30 min × $15/h. No bank pilot has measured it.

model-risk validation

the binding constraint on adoption — not compute, not tokens, not licences.

The ask: a 4-week pilot

- 50 anonymised real alerts with the target partner — MKBANK FCC (proposed)
- Three outcomes we commit to measure:
  - FPR at fixed recall, on real data
  - Minutes per alert, before vs after
  - Note-check pass rate on real cases
- Re-training is one script + a JSONL swap; entity rules and the verifier stay as-is

The analyst stays in the loop.

No external AI API. Synthetic demo data.

Live: **https://cornbread-left-licking.ngrok-free.dev** Repo: github.com/funstorm15/dalil · README + eval report + 91 tests

Thank you — questions?

synthetic demo · MKBANK: proposed pilot, not a confirmed partnership · metrics are our own benchmark — see README §5